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Stakeholder-informed renovation digital twin for multi-domain monitoring: Pilot implementation and acceptance evaluation

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ABSTRACT

Achieving the EU's energy and decarbonisation goals requires more efficient management of building renovation projects. Whilst digital twin (DT) technology is increasingly adopted across the built environment, most existing research and practice focus on its application during the operational phase of built assets such as energy monitoring, space and occupancy management, and predictive maintenance. However, there remains a gap in leveraging DT potential in renovation projects, supporting multi-domain performance tracking. This study presents a pilot implementation and preliminary evaluation of a Renovation Digital Twin (RDT) platform tailored to the specific challenges of renovation projects. The RDT integrates real-time data and dashboards to monitor 13 Key Performance Indicators (KPIs) across five domains: cost, schedule, quality, safety, and environment. Designed with GDPR compliance and usability in mind, the platform aims to support Architecture, Engineering, Construction, and Operations (AECO) stakeholders by improving transparency, early issue detection, and decision-making throughout project delivery. The RDT was evaluated using a quantitative questionnaire based on the Technology Acceptance Model (TAM) involving 102 industry professionals to assess perceived usefulness, ease of use, and overall user acceptance. Results demonstrated high user confidence, with *Attitude Toward Use* (RII = 0.894) ranking highest amongst TAM constructs. Deployment on the French RINNO demonstration site, achieving 73% emissions reduction but facing economic viability challenges, demonstrates the platform's capacity to surface complex multi-domain trade-offs. Whilst minor usability improvements were identified, initial findings provide evidence of the platform's viability and user acceptance, though broader validation across diverse renovation contexts is required. This research contributes an initial, stakeholder-informed DT framework for renovation, advancing the role of digital technologies in improving coordination, monitoring, and performance in occupied retrofit environments.

1. Introduction

The construction industry plays an essential role in global economic growth and the development of urban environments. However, the industry is considered the world's largest consumer of natural resources and a major contributor to environmental challenges such as resource depletion, carbon emissions, and energy use [1]. According to the 2019 Global Status Report for Buildings and Construction, the sector is responsible for 36% of final energy use and approximately 40% of energy- and process-related CO_2 emissions [2]. By 2045, the global urban population is projected to exceed six billion, potentially resulting in increased construction activities to meet housing and infrastructure demands [3]. In response to these environmental pressures, governments worldwide have implemented ambitious decarbonisation policies. The EU has established the Energy Performance of Buildings Directive (EPBD), setting energy savings targets for 2030 and 2033 [4], whilst the UK has committed to net-zero emissions by 2050, targeting carbon reductions of 68% by 2030 and 78% by 2035 [5]. Meeting these targets necessitates extensive building renovation: the UK alone requires retrofitting almost 29 million homes [6], exemplifying the scale of renovation needed globally to achieve climate goals [7].

However, achieving these ambitious renovation targets faces complex, multi-dimensional barriers. Understanding these implementation challenges requires recognising four critical barrier categories: human factors (social norms, awareness deficits, technology adoption resistance), technological factors (system integration challenges, interoperability issues), organisational factors (resource allocation, management support, professional training deficiencies), and external environment factors (regulatory complexity, policy inconsistencies, market constraints) [8]. This barrier framework demonstrates that technological solutions alone cannot address the low renovation rates constraining EU-wide climate targets. Instead, successful implementation requires

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integrated socio-technical approaches that simultaneously address technical functionality and human acceptance factors whilst coordinating diverse stakeholders with often conflicting motivations.

Digital twin (DT) technology offers a promising approach to overcoming these multi-dimensional barriers, provided it can effectively bridge the gap between technological capability and practical adoption. A DT represents an up-to-date virtual representation of a physical entity and its functional properties, leveraging dynamic data streaming and analytics capabilities to support informed decision-making [9,10]. DT applications in construction include optimised resource management through real-time monitoring [11]; enhanced stakeholder collaboration via shared digital platforms [12]; accelerated project delivery through virtual simulations and predictive analytics [13]; comprehensive progress monitoring using 4D BIM integration [14]; improved quality control and safety management [15,16]; and enhanced sustainability monitoring of carbon emissions and supply chains [17,18]. These capabilities position DT as particularly valuable for managing the complex, multi-domain aspects essential for successful renovation projects.

Despite this potential, applying DT technology in renovation contexts presents distinct challenges compared to new-build construction. New-build projects offer greater flexibility for technology integration without occupancy disruptions, whilst renovation projects must accommodate existing building constraints and occupied environments, significantly complicating DT implementation [19]. Current research demonstrates limited effective application of DT in renovation settings, with existing tools often inadequately adapted from new-build contexts [20]. Furthermore, most DT implementations focus on single-domain monitoring rather than the comprehensive, multi-domain approach required for managing renovation complexity, particularly during the onsite retrofitting phase where coordination challenges are most acute. This creates a critical research gap: while DT technology holds promise for addressing renovation barriers, its application for multi-domain performance monitoring during onsite renovation, including both technical implementation and user acceptance, remains underexplored.

This paper addresses this gap through a pilot study investigating the development, deployment, and evaluation of a Renovation Digital Twin (RDT) that enables multi-domain performance monitoring and delivers real-time feedback to project stakeholders. The research examines whether integrated DT platforms can effectively bridge technological capability with practical adoption in renovation contexts, evaluating both technical viability and user acceptance factors that influence technology diffusion. Using Design Science Research (DSR) methodology, the study grounds design requirements in stakeholder needs and evaluates platform performance through technical validation on a French demonstration site and quantitative user acceptance assessment with construction professionals. The paper proceeds as follows: Section 2 reviews related research; Section 3 presents the research methodology; Section 4 details RDT design, development, and deployment; Section 5 covers validation and evaluation; and Section 6 presents conclusions.

2. Literature review

2.1. DT Applications in building retrofit

Building retrofit currently relies on static tools with broad assumptions and generic inputs, often lacking digital modelling capabilities or methods for identifying optimal retrofit measures for specific projects [21,22]. DT technology offers potential to address these limitations through dynamically updated digital replicas for retrofit planning and decision-making. However, existing research reveals critical limitations in current DT implementations for renovation contexts. Most studies fail to address the multi-stakeholder, multi-domain complexity inherent in renovation projects, where coordination between occupants, contractors, and project managers becomes essential for success.

2.1.1. Project phases and DT coverage

Renovation projects involve distinct phases requiring different monitoring approaches and stakeholder coordination mechanisms. Design and planning phases (D&P) focus on scenario evaluation and retrofit strategy development. The onsite retrofitting phase (OR) involves active construction work where real-time monitoring becomes critical for managing quality, safety, cost, and schedule performance. Operation and maintenance phases (O&M) address post-renovation building performance optimisation. Current DT research shows uneven distribution across these phases. Most studies address design/planning activities or post-renovation operations, whilst the onsite retrofitting phase, where coordination challenges are most acute, receives limited attention. This gap is particularly significant given that renovation projects typically experience greater time and cost overruns, increased safety risks, and quality issues compared to new-build construction [23].

2.1.2. Use cases and applications

Building on Pan et al.'s [24] theoretical framework for DT in renovation, current applications have emerged across four primary domains. **Indoor condition monitoring** maintains occupant health and safety during renovation through environmental parameter tracking. Arsiwala et al. [25] developed a DT solution integrating IoT, BIM, and AI to automate CO_2 emission monitoring, demonstrating spatial data visualisation for air quality management with machine learning-based emission prediction. Studies by Mainini et al. [26], Daniotti et al. [27], and Banfi et al. [28] developed systems for monitoring temperature, humidity, and CO_2 emissions. Tan et al. [29] proposed a DT lighting system for dynamic occupant comfort management. **Energy modelling and monitoring** represents the most developed application area, reflecting retrofitting's energy efficiency goals. Jradi et al. [22] proposed *DanRETwin* for decision-making and energy optimisation in existing buildings through IoT sensors and AI-driven performance analysis. Bocullo et al. [30] developed a city block renovation approach using renewable energy sources with scenario evaluation capabilities. Trombadore et al. [31] and Kaewunruen et al. [32] focused on energy modelling for retrofit scenario evaluation. **Cost analysis applications** address economic evaluation during planning phases. Trombadore et al. [31] developed a methodology for heritage building renovation, providing tools to evaluate cost-effectiveness whilst monitoring air quality, building performance, and user behaviour. **Safety monitoring** remains the least developed domain. Khajavi et al. [33] developed a DT architecture emphasising real-time fire detection, incorporating data fusion, localisation, and propagation simulation. Their approach demonstrated safety benefits during renovation when building systems are offline, though it addressed only fire hazards.

2.1.3. Technical components

DT implementations integrate four core technological components: data sensing devices for reality capture and performance tracking; BIM and 3D visualisation for digital representation; visualisation dashboards for stakeholder communication; and AI for automated data processing. Most studies rely on IoT sensors for data exchange between physical and digital systems, collecting parameters including temperature, humidity, occupancy, luminance, CO_2, and energy consumption [22,27]. However, BIM capabilities remain underutilised, with most studies focusing solely on 3D visualisation rather than leveraging design coordination, clash detection, or 4D simulation functions [34]. Visualisation dashboards are underdeveloped, with limited implementations providing comprehensive monitoring across multiple performance domains. AI adoption remains restricted to specific applications such as emission prediction and computer vision for hazard detection [25,33].

2.2. Technology acceptance frameworks

The Technology Acceptance Model (TAM), developed by Davis [35], provides a framework for understanding technology adoption through

Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitude Toward Use (ATU), and Behavioural Intention (BI). In construction contexts, studies of BIM technology acceptance demonstrate that perceived usefulness typically emerges as the primary adoption driver [20]. However, the complexity of occupied renovation environments may alter traditional TAM relationships, potentially emphasising attitude toward use where trust and organisational alignment become critical factors. Limited research has addressed DT technology acceptance in construction contexts [36]. Most validation approaches focus on technical functionality testing without systematic user acceptance assessment, creating a gap in understanding how human factors influence DT adoption in multi-stakeholder renovation environments.

2.3. Research gaps and contributions

Table 1 reveals three critical gaps in existing DT research for renovation contexts. First, project phase analysis demonstrates uneven research distribution with only 27% of studies targeting onsite retrofitting activities where coordination challenges are most acute. Among onsite-focused studies, only the current RDT platform provides comprehensive multi-domain performance monitoring. Second, most studies focus on single-domain monitoring rather than the multi-domain approach required for managing renovation complexity. Only one previous study attempted triple-domain monitoring [31], whilst none addresses all five performance domains simultaneously. Third, systematic assessment of DT acceptance using established theoretical frameworks remains limited. This gap is significant given that technology adoption in renovation projects depends on alignment with diverse organisational needs and stakeholder motivations beyond technical functionality. The contribution of this research is threefold: (i) present a pilot study of a comprehensive RDT solution enabling multi-domain monitoring during onsite retrofitting activities; (ii) provide detailed methodology for DT implementation in renovation contexts; and (iii) evaluate the platform using TAM methodology, addressing the gap in technology acceptance evaluation for multi-domain platforms in renovation environments.

3. Research methodology

Given the identified gap in onsite retrofitting research and the need for multi-domain performance monitoring (Section 2.3), this study adopts a DSR methodology to develop and evaluate a comprehensive RDT platform. The research addresses the theoretical challenge of bridging DT technological potential with practical adoption in renovation contexts, specifically targeting the under-researched onsite retrofitting phase where coordination challenges are most acute. The RDT framework is part of the Horizon Europe-funded RINNO research initiative, which aims to optimise deep renovation processes across Europe [38]. The RINNO project involves 19 partners from 10 EU countries with 4 demonstration sites, providing a robust collaborative environment for developing and testing renovation technologies. The French demonstration site serves as the primary case study for the RDT implementation [39].

3.1. Research questions

The research builds upon the comprehensive barrier framework established in the foundational RINNO study, which identifies four critical categories constraining deep renovation efforts: human, technological, organisational, and external environment barriers [8]. This framework reveals that effective technologies alone cannot solve low renovation rates, necessitating integrated solutions that address both technical functionality and human acceptance factors. Building on this framework and the identified research gaps (Section 2.3), this study addresses two primary research questions:

RQ1 - Platform development and deployment: can a multi-domain DT platform be developed that integrates real-time performance data across cost, schedule, quality, safety, and environmental domains whilst maintaining GDPR compliance and functional operation in occupied renovation environments? This question addresses the technical feasibility gap identified in the literature review, where existing DT implementations rarely target onsite retrofitting with comprehensive multi-domain monitoring (Table 1).

RQ2 - User acceptance evaluation: what are construction professionals' perceptions of the RDT platform's usefulness, ease of use, attitude toward using the system, and intention to adopt it in future renovation projects? This question addresses the human acceptance factors critical for DT adoption in multi-stakeholder renovation environments, evaluated through the TAM framework. Given the RINNO barrier framework's emphasis on human factors as equal to technological barriers [8], understanding user acceptance is essential for assessing the platform's practical viability beyond technical functionality.

3.2. DSR Implementation

The methodological approach follows the five-stage DSR process proposed by Peffers et al. [40], with each stage addressing specific aspects of the research challenge (Fig. 1):

Stage 1 – Problem identification and motivation: Literature review identified research gaps in onsite retrofitting monitoring and multi-domain performance tracking. Collaborative workshops with fourteen industry stakeholders from RINNO consortium partners, including onsite workers, site managers, project managers, and engineers, identified information gaps, reporting needs, and workflow bottlenecks during retrofitting activities.

Stage 2 – Definition of objectives: Stakeholder input was categorised into five key use cases: quality, cost, scheduling, safety, and environment. Through structured workshops, 13 critical KPIs were identified and defined according to four criteria: relevance to renovation project success, feasibility of data collection in occupied environments, alignment with industry practices, and integration potential with digital monitoring systems. Each KPI includes specific data sources, calculation formulas, and visualisation formats (Table 2). Building on stakeholder input from Stage 1, weekly measurement frequencies were established to balance data granularity with practical site reporting capabilities.

Stage 3 – Design and development: The RDT platform was developed through iterative prototyping. An initial version was deployed locally via NGROK [41] for stakeholder testing and feedback. The final platform, incorporating BIM and comprehensive dashboards, was deployed on HEROKU cloud platform [42] for scalable access.

Stage 4 – Demonstration and evaluation: The RDT was validated through technical testing using the French demonstration site BIM model and functionality verification, followed by TAM-based user acceptance evaluation with 102 industry professionals.

Stage 5 – Communication: Research outcomes, including design artefacts, empirical findings, and methodological insights, are reported in this paper with contributions to theory and practice.

4. The RDT platform

The RDT platform is designed to support the monitoring of retrofitting progress and the timely provision of feedback to project stakeholders. Its main objectives, informed by findings from the literature review and stakeholder input, are to: (i) collect onsite information related to weekly project performance, including safety, quality, cost, task completion, delays, and waste management; and (ii) provide stakeholders with timely insights to support informed decision-making and corrective actions when necessary.

Table 1
Comparative analysis of DT technology applications in renovation projects.

Study	Purpose of the Study	Use Cases				Components				Project Phase	Testing/ Evaluation
		Indoor condition	Energy	Cost	Safety	Data Sensing	BIM	Dashboard	AI		
Mainini et al. [26]	Develops BIM-based tools 1) to improve data sharing on construction work progress among renovation stakeholders, and 2) to monitor indoor conditions, comfort preferences and energy consumption with inhabitants.	✓	✓			✓	✓	✓		OR	
Arsiwala et al. [25]	Automates the monitoring and controlling of equivalent CO_2 emissions through integrating IoT, BIM and AI.	✓				✓	✓	✓	✓	O&M	✓
Jradi et al. [22]	Optimises energy retrofits, retro-commissioning and data-driven performance optimisation of non-residential existing buildings.	✓	✓			✓			✓	D&P	
Bocullo et al. [30]	City block renovation using renewable energy sources like solar PV, heat pumps, and electric heaters.		✓	✓			✓			D&P	
Khajavi et al. [33]	Safety DT, focusing on early fire and anomaly detection, fire localisation, intensity evaluation, and sharing real-time incident data.				✓	✓	✓	✓	✓	OR	✓
Daniotti et al. [27]	Develops a BIM-based toolkit for enabling efficient renovation of residential buildings.	✓				✓	✓			D&P	
Banfi et al. [28]	Proposes a DT and Cloud BIM-XR platform to improve efficiency and comfort of existing buildings through retrofits.	✓	✓			✓	✓	✓		O&M	
Hou et al. [37]	Building envelope energy audits and simulations, with traceability of retrofit plans back to the envelope components.		✓			✓	✓			D&P	
Tan et al. [29]	Develops an intelligent indoor lighting system and reducing operation and maintenance costs using BIM and computer vision.	✓		✓		✓	✓		✓	O&M	✓
Trombadore et al. [31]	Evaluates retrofitting scenarios, decision-making on cost-effective and innovative energy rehabilitation actions.	✓	✓	✓		✓	✓			D&P	
Kaewunruen et al. [32]	Evaluates Net-Zero Energy Buildings (NZEB) for existing buildings and applying renewable technologies.		✓				✓			D&P	
Current Study RDT	Multi-domain performance monitoring for renovation project management with comprehensive KPI tracking and user acceptance evaluation	✓	✓	✓	✓	✓	✓	✓		**OR**	✓

Legend: D&P = Design and Planning; OR = Onsite Retrofitting; O&M = Operation and Maintenance

4.1. Design requirements

The development of the RDT platform is grounded in a set of design requirements (DR), including key components and high-level functionalities, identified in collaboration with the RINNO project's industrial partners. As outlined in Table 3, seven core requirements must be fulfilled to develop a DT platform suited to the specific needs of renovation projects.

The RDT platform implements role-based authentication to ensure data security, GDPR compliance, and appropriate stakeholder collaboration during the onsite retrofitting activities. User authentication employs industry-standard protocols including OAuth 2.0, OpenID Connect

Table 2
Renovation project KPIs with definitions and measurement protocols.

Domain	ID	Name	Description	Data Source and Calculation	Visualisation
Quality	1	Quality incidents	Count of quality incidents requiring corrective action	Mobile app input by site managers via standardised incident recording forms	Line chart
	2	Quality controls monitoring	Upcoming quality inspections scheduled within the next 10 days	API integration with project schedule and quality management system	List format
	3	Customer complaints	Count of formal complaints raised by occupants or clients	API integration with complaint database and quality management system	Line chart
	4	Recurring quality issues alerts	Identification and ranking of the 5 most frequent quality problems	Automated frequency analysis of quality incident database	Ranked list
Cost	5	Cost savings	Cumulative monetary savings achieved against baseline budget	API integration with project financial database and cost management system	Line chart
	6	Cost overruns	Total expenditure exceeding approved budget baseline	API integration with project financial database and cost management system	Line chart
Scheduling	7	Delay monitoring	Variance between actual days worked and planned schedule	Mobile app reporting of actual progress compared with planned schedule	Line chart
	8	Milestones monitoring	Percentage distribution of achieved, ongoing, and upcoming project milestones	API integration with project schedule database and milestone tracking system	Pie chart
	9	Issue resolution duration	Average time elapsed between issue opening and closure	Automated calculation from issue management database timestamps	Line chart
Safety	10	Recurring safety issues alerts	Identification and ranking of the 5 most frequent safety hazards	Automated frequency analysis of safety incident database	Ranked list
	11	Safety incidents	Count of recorded safety events including near-misses and accidents	Mobile app reporting with severity classification integrated with safety database	Line chart
	12	Safety incident control monitoring	Stakeholders and organisations involved in safety issue management and resolution	Safety database integration with stakeholder assignment and responsibility tracking	List format
Environment	13	Waste monitoring	Quantity of waste generated measured by number of recorded waste containers	Mobile app logging of waste container collections with waste type categorisation	Line chart

Table 3
Design requirements of the RDT platform.

ID	Description
DR1	Provides regular updates on relevant project KPIs related to quality, cost, scheduling, safety, and environment (see Table 2).
DR2	Stores historical project KPI data throughout the retrofitting phase.
DR3	Supports the simulation of project KPIs and their progression over time.
DR4	Offers a user-friendly dashboard that integrates BIM models with relevant visual representations.
DR5	Notifies stakeholders when project KPIs exceed predefined thresholds.
DR6	Ensures remote access to the RDT platform at any time and from any location via the internet for all project stakeholders.
DR7	Complies with applicable regulations, such as the General Data Protection Regulation (GDPR), when collecting and processing building data.

Table 4
RDT platform user roles and core access permissions.

User role	Visualize current KPIs	Access historical data	4D BIM simulation	Configure KPI thresholds
Project owner	✓	✓	–	–
Construction director	✓	✓	✓	✓
Construction site manager	✓	✓	✓	✓
Construction team member	✓	✓	–	–
Site supervisor	✓	✓	✓	✓
Site worker	✓	✓	–	–
Subcontractor	–	–	–	–
Architect	–	–	–	–
Design coordinator	–	–	–	–
Design manager	✓	✓	✓	–
Design team member	✓	✓	✓	–

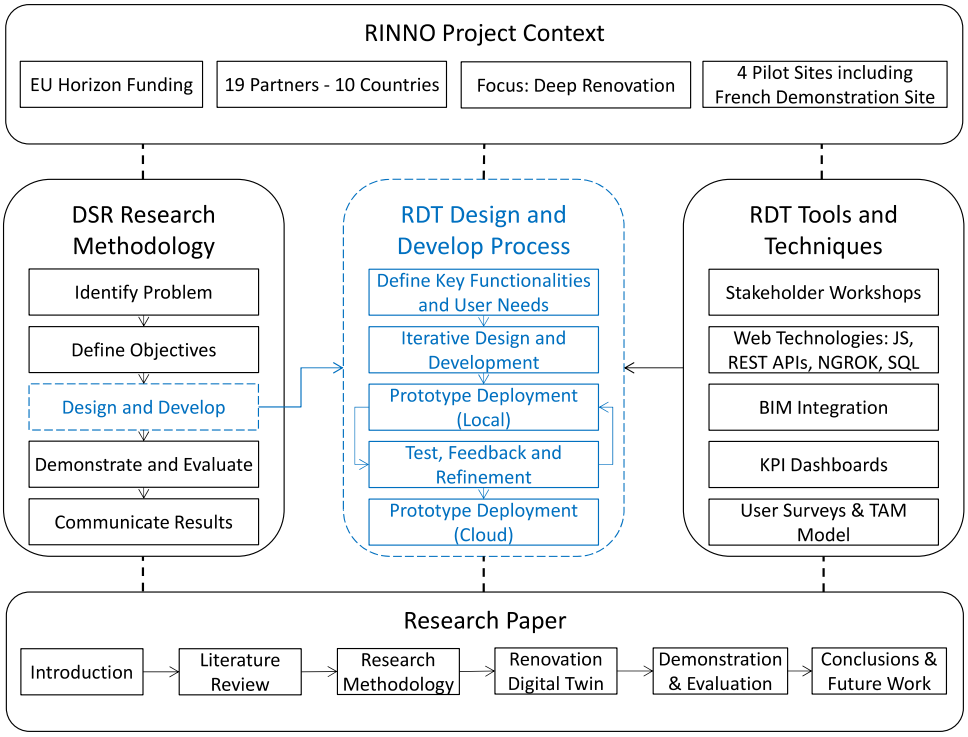


Fig. 1. Research methods, tools and techniques (black) and RDT platform design and development process (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

(OIDC), and Security Assertion Markup Language (SAML), enabling login through multiple credential sources. New users register via Email verification, after which platform administrators validate accounts and assign appropriate roles based on project responsibilities. Through collaborative workshops with industrial partners of the RINNO consortium, eleven distinct user roles were identified, each with specific access rights aligned to typical hierarchies within renovation projects (Table 4).

4.2. System architecture

The development of the RDT platform is guided by the seven-component architecture presented in Fig. 2. This architecture addresses key challenges in DT implementation for renovation contexts: multi-domain data integration, stakeholder-specific access control, real-time performance feedback, and GDPR compliance. The dual-server design separates raw data from processed analytics, enabling both real-time monitoring and historical analysis whilst maintaining data integrity. Each component fulfils specific design requirements identified in Table 3.

The *Acquisition Service* collects raw data from onsite renovation activities (quality incidents, complaints, safety issues) and transmits them to the *Data Storage Server*. Data collection employs two primary channels: (i) manual input via mobile application for quality, safety, and environmental metrics, and (ii) API integration with existing project management systems for cost and schedule data. To ensure GDPR compliance (DR7), data collection is implemented through these non-intrusive methods rather than fixed sensing technologies (e.g. IoT devices or sensors), which typically require occupant and operator consent [43]. Project managers use the mobile application to input data related to project execution and progress, which is also shared with traditional project management software such as Teamoty, SAP, and Dalux to enable integration with existing project workflows. API connections utilise RESTful endpoints with JSON data exchange formats, whilst data transmission employs secure HTTPS protocols. Data synchronisation occurs on a weekly basis for systematic metrics (cost, schedule, quality controls) with event-driven reporting for critical incidents (safety events, quality issues, complaints), enabling convergence between physical and digital onsite states at a rate appropriate to renovation project needs [44]. The *Data Storage* component consists of a Structured Query Language (SQL) server that stores the collected raw data [45], serving as the primary data source for the *Processing Service* (DR1). The *Processing Service* retrieves raw data from Server 1 and queries traditional project management software via API to collect supplementary schedule, cost, and safety data. The service computes relevant KPIs using SQL queries and stores the output in *KPI Storage* (Server 2), generating reliable KPI data for downstream services and ensuring comprehensive multi-domain monitoring (DR1). The *Notification Service* functions as a continuous background process (the 'KPI watcher') that monitors KPI values and triggers alerts when predefined thresholds are exceeded. For each KPI (Table 2), minimum values ensure data quality and consistency, whilst maximum values trigger warnings visualised through colour-coded indicators in the *Visualisation Service* (DR5). The *Simulation Service* enables time-based simulation of historical KPI values stored in Server 2, visualised using a 'Timeliner' module integrated into the *Visualisation Service*. KPI trends are mapped onto BIM model elements using Forge APIs [46], supporting 4D simulation [43] (DR3). The *KPI Storage* component, hosted on a second SQL server, maintains calculated KPIs and serves as the central data source for the *Simulation*, *Notification*, and *Visualisation* services (DR2). The *Visualisation Service* provides the central interface through which users access all visual content, including current and historical KPIs. It integrates the BIM model via Forge APIs and presents data through multiple formats: shaded BIM elements, 4D timelines, and domain-specific charts and graphs (Figs. 3 and 4). This component offers an intuitive and information-rich dashboard (DR4).

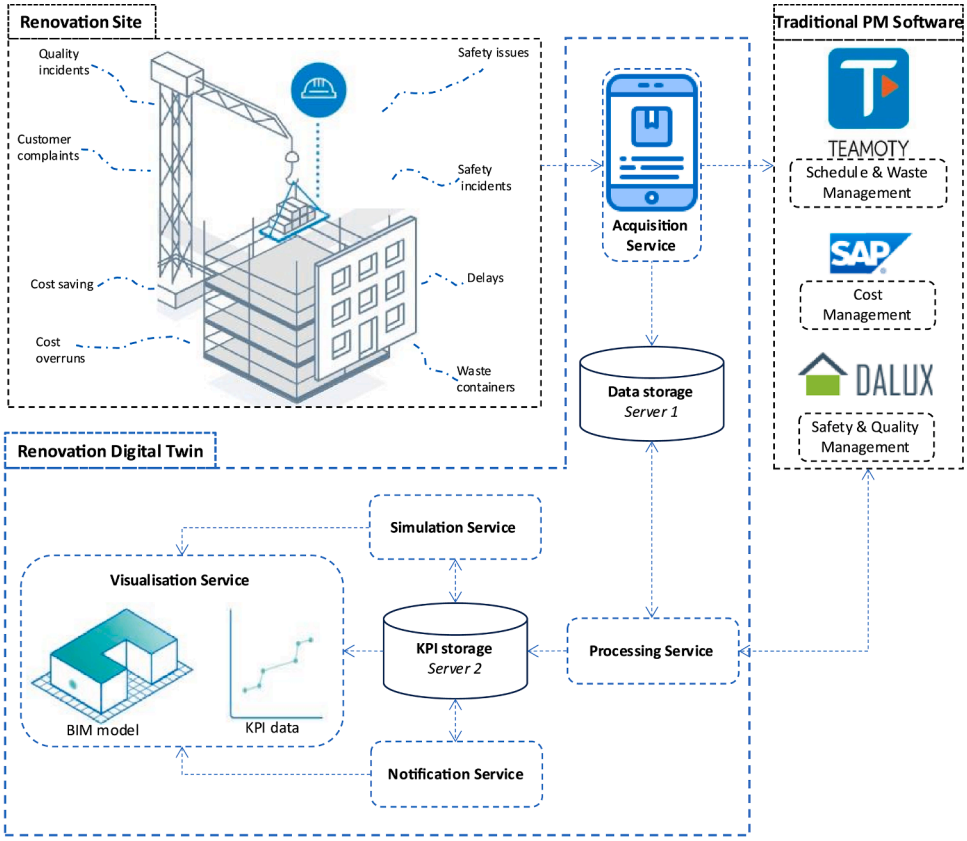


Fig. 2. RDT system architecture and components.

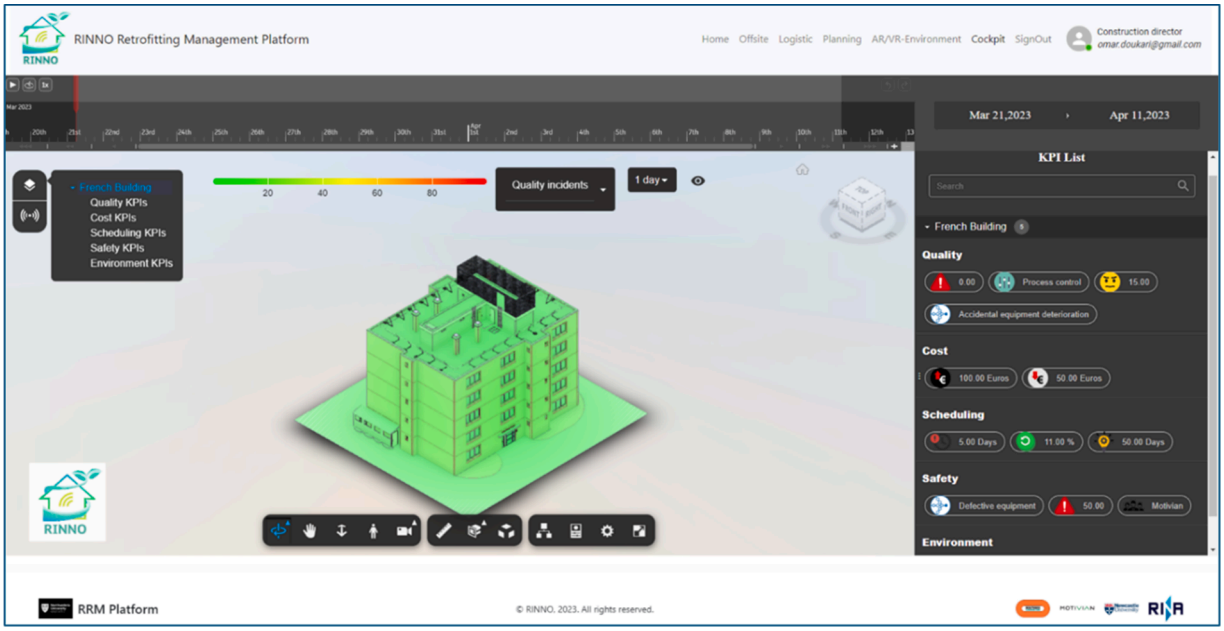


Fig. 3. RDT main user interface.

As a web application, the RDT ensures anytime-anywhere access for all project stakeholders via the internet (DR6).

The RDT platform was developed using JavaScript for the backend and the React library [47] with CSS and HTML for the frontend interface. The MUI library [48] was employed to ensure simplicity, clarity, and UI responsiveness across various device types (smartphones, tablets, laptops) in both onsite and offsite contexts.

4.3. Platform deployment

4.3.1. Demonstration site selection

The RDT platform was deployed and evaluated using the French demonstration site within the RINNO project (Project Sarrazins, Lille). The building is a 4-level residential structure comprising 29 small apartments (T1) undergoing comprehensive energy retrofit. The renovation

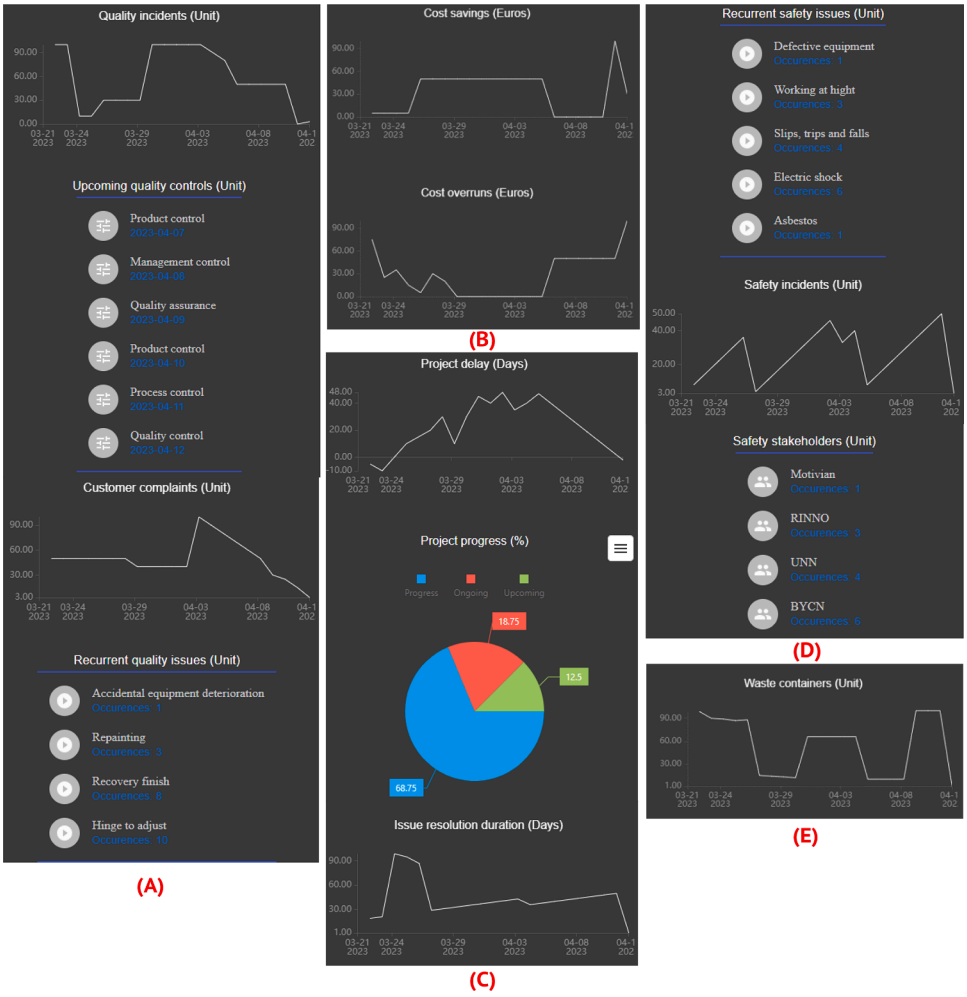


Fig. 4. RDT platform KPI dashboard views for: (A) quality, (B) cost, (C) scheduling, (D) safety, and (E) environmental performance.

included external insulation with two layers, high-performance double glazing windows with roller shutters, rooftop photovoltaic panels, and an Inventilate ventilation system with advanced HVAC equipment. The project used innovative robotic facade insulation techniques whilst maintaining occupancy during construction (Fig. 5) [39].

4.3.2. Deployment architecture

The RDT deployment followed a two-step process. First, for testing purposes, the platform was deployed on a local machine using NGROK [41] and made accessible to the RINNO project's industrial partners to demonstrate its design and validate its core functionalities. Second, the platform was permanently deployed via HEROKU, a cloud-based service that supports the hosting and publication of third-party web applications [42]. A brief demonstration video showcasing the RDT platform in action is available at [49].

4.3.3. Sustainability performance context

The lifecycle assessment of the French demonstration site provides empirical context for understanding the renovation challenges the RDT is designed to address. Analysis over a 50-year reference period following the Level(s) EU Framework [50] revealed the performance outcomes summarized in Table 5 [51]. Environmental performance improved substantially: primary energy demand decreased 46% (from 252,827 to 136,171 kWh annually) and greenhouse gas emissions fell 73% (from 26,458 to 7,033 kgCO_2-eq annually). However, lifecycle costs increased 39% (from €16,879 to €23,389 annually), with €753,000 capital investment yielding negative financial returns (NPV: -€326,000; IRR: 1.9%; no payback within the 50-year analysis period). The high costs were driven primarily by regulatory requirements for brick façade finishing and advanced HVAC systems.

Table 5
French pilot site renovation performance summary (50-year lifecycle analysis).

Performance Indicator	Baseline	Post-Retrofit	Change
Annual Primary Energy (kWh)	252,827	136,171	−46%
Annual GHG Emissions (kgCO_2-eq)	26,458	7033	−73%
Annual Lifecycle Cost (€)	16,879	23,389	+39%

5. Demonstration and evaluation

The RDT was validated through a two-phase approach: (i) technical testing using a proof-of-concept implementation on the French demonstration site within the RINNO project, and (ii) user acceptance evaluation through a questionnaire survey with construction and retrofitting practitioners. The objective was to verify technical functionality and system integration whilst assessing whether the platform meets end-user requirements and expectations.



Fig. 5. French pilot site: before (left) and after (right) deep renovation.

5.1. Technical validation

The RDT platform's requirements and functionalities (Table 3) were tested using the BIM model of the French building described in Section 4.3. The evaluation employed React Developer Tools [52], a Chrome DevTools extension, to systematically verify system components and ensure error-free operation across all seven RDT services. Testing confirmed successful calculation and display of all 13 KPIs, validated API communications through console outputs, verified data integration protocols, and demonstrated stable operation across different user roles. The platform's technical validation focused on functional verification rather than performance measurement. All KPI calculations were verified against their defined formulas (Table 2), threshold monitoring functioned correctly with colour-coded alerts, and historical data storage enabled trend analysis through the 4D simulation interface. Testing confirmed seamless integration with the RINNO Suite through successful JSON data exchange protocols across all KPI categories [53]. Performance measurement on live projects were identified as future research directions.

5.2. User acceptance study

User acceptance was assessed using the TAM model [35] across four constructs: *Perceived Usefulness* (PU), *Perceived Ease of Use* (PEOU), *Attitude Toward Use* (ATU), and *Behavioural Intention* (BI). A 24-item questionnaire (5-point Likert scale: 1 = strongly disagree, 5 = strongly agree) was distributed to AECO professionals. The instrument comprised 6 PU questions measuring productivity enhancement (retrofitting efficiency, project outcomes, decision support, safety improvements, real-time feedback, overall platform value), 10 PEOU questions assessing usability dimensions (training needs, interface design, learning curve, technical accessibility, data retrieval, system integration, information clarity, update mechanisms, visual presentation, independent operation), 4 ATU questions evaluating user perceptions (design satisfaction, purpose confidence, feature value, needs alignment), and 4 BI questions measuring adoption commitment (future implementation, peer endorsement, feature engagement, sustained usage) (see Fig. 6).

Purposive sampling targeted diverse AECO roles, including professionals without DT experience. Data analysis employed descriptive statistics, Cronbach's Alpha reliability testing (α), and Relative Importance Index: $\text{RII} = \frac{5N_5+4N_4+3N_3+2N_2+1N_1}{A\times N}$, where N_5–N_1 represent response frequencies, A is the highest weight (5), and N is total respondents. Statistical significance was tested using one-sample t-tests against the neutral response (3.0) with $\alpha = 0.05$. A total of 102 responses were collected over 6 weeks from site engineers (34.3%), architects (23.5%), project managers (10.8%), contractors (8.8%), civil engineers (7.8%), consultants (5.9%), BIM managers (4.9%), and BIM coordinators (3.9%). Experience levels comprised 72.5% with less than 5 years, 19.6% with 5–10 years, 3.9% with 10–15 years, and 3.9% with more than 15 years. Gender distribution was 57.7% male, 26.9% female, and 15.4% unspecified.

5.3. Results and discussion

Cronbach's Alpha confirmed strong internal consistency across all constructs: PU ($\alpha = 0.854$), PEOU ($\alpha = 0.857$), ATU ($\alpha = 0.865$), and BI ($\alpha = 0.896$), all exceeding the 0.7 threshold. All construct means significantly exceeded the neutral midpoint (3.0) at $p < .001$, indicating positive user perceptions across all TAM dimensions.

5.3.1. TAM Construct performance

Fig. 6 illustrates positive responses across all constructs. PU scores ranged from 4.24 to 4.49 ($M = 4.34$, $SD = 0.69$), with the highest score for 'Overall Platform Value' ($M = 4.49$) and lowest for 'Decision Support' ($M = 4.24$). PEOU scores ranged from 4.10 to 4.57 ($M = 4.35$, $SD = 0.64$), with the highest score for 'System Integration' ($M = 4.57$) and lowest for 'Independent Operation' ($M = 4.10$). ATU demonstrated the highest overall performance, ranging from 4.33 to 4.61 ($M = 4.47$, $SD = 0.57$), with both 'Purpose Confidence' and 'Needs Alignment' achieving $M = 4.61$. BI ranged from 4.25 to 4.41 ($M = 4.34$, $SD = 0.66$).

5.3.2. Relative importance rankings

RII analysis revealed ATU as the strongest adoption driver (RII = 0.894), followed by PEOU, PU, and BI (all 0.87). Three performance tiers emerged across the 24 survey items: *Top tier* (RII $\geq$ 0.90): 'Purpose Confidence' and 'Needs Alignment' (both RII = 0.92, rank 1), 'System Integration' (RII = 0.91, rank 3), 'Overall Platform Value' (RII = 0.90, rank 4), and 'Training Needs' (RII = 0.90, rank 5). This tier demonstrates that conceptual alignment and system interoperability are primary adoption drivers. *Middle tier* (0.85 $\leq$ RII < 0.90): Most operational aspects including interface design, learning curve, information clarity, visual presentation, and functional usefulness dimensions, representing satisfactory but improvable features. *Lower tier* (RII < 0.85):

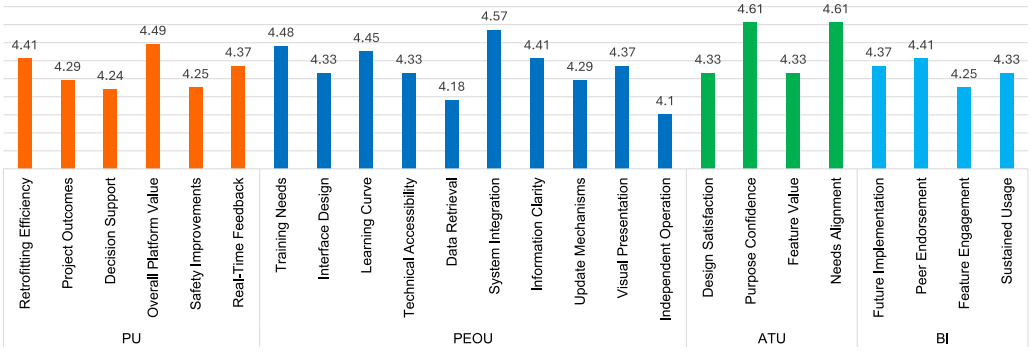


Fig. 6. Mean scores for individual TAM survey items grouped by construct: Perceived Usefulness (orange), Perceived Ease of Use (blue), Attitude Toward Use (green), and Behavioural Intention (light blue) (n = 102). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

'Decision Support' (RII = 0.85, rank 22), 'Data Retrieval' (RII = 0.84, rank 23), and 'Independent Operation' (RII = 0.82, rank 24), identifying specific improvement priorities.

5.3.3. Discussion

The TAM evaluation reveals comparable importance across all constructs, with attitude toward use highest (RII = 0.894), followed by perceived ease of use, perceived usefulness, and behavioural intention (all RII = 0.87). The narrow range (0.87–0.894) suggests that successful DT adoption requires addressing multiple dimensions simultaneously rather than prioritising any single factor. This finding challenges simplified technology acceptance models that emphasise usefulness or ease of use alone, and aligns with the RINNO framework's emphasis on comprehensive stakeholder engagement [8].

Analysis of individual items reveals a notable performance gap: Purpose Confidence and Needs Alignment (both RII = 0.92, rank 1) exceed Independent Operation (RII = 0.82, rank 24). This 11% difference indicates that whilst users exhibit strong conceptual acceptance of the platform's value and relevance, they demonstrate lower confidence in autonomous usage. This pattern suggests that organisational factors, such as training provision, technical support availability, and institutional implementation strategies, may critically influence adoption outcomes. The predominance of early-career professionals (72.5% with less than 5 years experience) in the sample may contribute to lower independent operation confidence scores, as professional experience typically correlates with technology self-efficacy in construction contexts [54]. This demographic pattern requires further investigation through experience-stratified analysis.

System Integration (RII = 0.91, rank 3) emerged as the highest-ranked ease of use factor, demonstrating that perceived interoperability with established BIM tools and workflows facilitates acceptance. This finding supports design principles emphasising seamless integration over standalone systems [55]. The difference between attitude toward use (RII = 0.894) and behavioural intention (RII = 0.87) may indicate that external contextual factors, such as organisational policies, budget constraints, or project-specific requirements, influence implementation decisions beyond individual user acceptance. Future research should examine these organisational and contextual moderators through longitudinal adoption studies.

The finding that attitude toward use achieved the highest acceptance score, albeit marginally, suggests that in multi-stakeholder renovation environments, trust and organisational alignment may be as important as functional performance or ease of use. This aligns with the RINNO barrier framework's identification of human factors (trust, alignment) as equally critical as technological factors (functionality, usability) in determining adoption outcomes [8]. The stakeholder-informed design process, which positioned users as active co-creators throughout development, likely contributed to the strong conceptual acceptance reflected in high Purpose Confidence and Needs Alignment scores.

5.3.4. RDT Application context

The French demonstration site's lifecycle assessment (Section 4.3, Table 5) illustrates the complex sustainability landscape the RDT is designed to navigate. The 73% emissions reduction achieved substantial environmental goals, yet the 39% cost increase (NPV: -€326,000, no payback) reveals financial viability challenges that require transparent multi-stakeholder communication, precisely the coordination need that motivated high ATU scores (RII = 0.894). This tension between environmental imperatives and economic constraints exemplifies why multi-domain KPI tracking becomes critical in occupied retrofit environments. The RDT's simultaneous monitoring of cost, schedule, quality, safety, and environmental performance enables stakeholders to identify trade-offs in real-time, supporting the informed decision-making necessary when capital costs are high and regulatory requirements strict. Technology-specific insights from the assessment (e.g., biobased K-FLEX insulation achieving comparable thermal performance whilst reducing embodied emissions) demonstrate the granular visibility the platform provides. The 50-year lifecycle perspective captured in Table 5 reveals how operational savings offset upfront environmental impacts, insights that would support evaluating policy interventions (subsidies, incentives, carbon pricing) required to bridge economic viability gaps. This contextualizes the platform's value proposition: in projects where environmental performance is technically achievable but economically contingent on supportive frameworks, the RDT's multi-domain data becomes essential for stakeholder alignment and evidence-based policy advocacy.

5.4. Limitations and future work

Several considerations should be acknowledged when interpreting these findings and planning future implementations. The RDT platform was tested on a single renovation project (Project Sarrazins, Lille), limiting generalisability to other retrofit contexts involving different building types, scales, and renovation scopes. The proof-of-concept demonstration focused on functional verification rather than comprehensive

empirical validation. Whilst the TAM-based evaluation provided insights into user acceptance, the study lacks quantified performance improvements (cost savings, schedule adherence, safety incident reduction) that could validate the platform's core benefits. The short evaluation period may not capture long-term usability issues or effectiveness throughout complete renovation lifecycles. The TAM sample exhibited bias toward early-career professionals (72.5% with less than 5 years experience), potentially limiting applicability to experienced practitioners with different adoption patterns. Technical limitations include absence of AI capabilities, with the simulation service supporting only historical KPI visualisation without predictive analytics for proactive issue identification. Reliance on manual data entry through mobile applications may introduce inconsistencies compared to automated sensor-based collection systems. The evaluation relied on self-reported perceptions rather than observed usage behaviours, which may not accurately reflect actual adoption patterns in real-world scenarios [56]. The sustainability outcomes from the French demonstration site (Section 4.3, Table 5) represent a single case study with specific economic and regulatory characteristics that may limit generalisability of the cost-benefit findings.

Future research should address these considerations through multi-site validation across diverse renovation projects (residential versus commercial, varying scales and types) in different climate zones and regulatory environments to establish broader applicability. Longitudinal performance studies tracking actual project metrics before, during, and after RDT implementation would provide empirical evidence of platform effectiveness. Research involving experienced professionals and comparative analysis between RDT-supported projects and traditional approaches would quantify relative benefits and identify optimal implementation strategies. Technical enhancements should prioritise automated data integration through IoT sensors alongside predictive analytics using deep learning for proactive project management support [57]. Integration of predictive monitoring methods into the RDT platform to assess the impact of renovation works on indoor conditions could inform occupants and stakeholders before retrofitting processes begin [58]. Likewise, incorporation of material circularity metrics into the RDT platform to monitor reuse rates, recycled content percentages, and end-of-life recoverability would align renovation tracking with EU Circular Economy Action Plan targets [59]. The current RDT framework provides foundation for developing cognitive digital twin systems capable of learning from project patterns, providing intelligent recommendations, and supporting autonomous decision-making processes [60].

6. Conclusions

This paper presents the development, deployment, and evaluation of an RDT platform designed to support multi-domain performance monitoring in building retrofit projects. The research addresses identified gaps in DT applications for renovation contexts, where existing frameworks predominantly focus on operational phase monitoring rather than onsite construction performance across quality, cost, schedule, safety, and environmental domains. The RDT was developed using DSR methodology, incorporating stakeholder input from RINNO consortium partners to define 13 KPIs across five performance domains. Technical validation on the French demonstration site confirmed functional operation across all seven system components, successful BIM integration, multi-stakeholder access control, and GDPR compliance through mobile-based data collection.

User acceptance evaluation with 102 construction professionals revealed strong positive perceptions across all TAM constructs. Attitude toward use emerged as the strongest adoption driver (RII = 0.894), with Purpose Confidence and Needs Alignment as highest-rated items (both RII = 0.92), demonstrating strong conceptual acceptance of the platform's value and relevance. System Integration (RII = 0.91) emerged as the most valued ease of use factor, whilst Independent Operation (RII = 0.82) identified training and support as priority enhancement areas. These findings confirm that interoperability with established BIM workflows critically facilitates adoption [55].

Theoretically, the research extends TAM to multi-stakeholder construction environments, revealing that attitude-driven acceptance may be as important as perceived usefulness in complex renovation contexts. This finding provides empirical support for the RINNO barrier framework [8], demonstrating that human barriers (trust, alignment) can equal technological barriers (functionality, usability) in determining adoption outcomes. The study advances socio-technical systems theory by demonstrating successful coordination of multiple performance domains through GDPR-compliant architecture with stakeholder-specific access control.

Practically, the stakeholder-informed design process positioned users as active co-creators, embodying the 'human-in-the-loop' principle essential for effective DT implementation [61]. Strong performance of integration-related factors provides empirical guidance emphasising that seamless workflow integration represents a critical success factor. The platform demonstrates potential for establishing new project management routines supporting sustainability's three pillars: enhanced collaboration (social), waste reduction (environmental), and improved cost and schedule management (economic). Deployment on the French demonstration site illustrates this multi-domain capability: whilst achieving substantial environmental performance (73% emissions reduction), the project faced significant economic challenges (39% cost increase, NPV: -€326,000), exemplifying the complex trade-offs the RDT's KPI tracking framework is designed to make transparent to stakeholders. Successful DT implementation requires not only technical capability but also systematic attention to human factors, training provision, and organisational change management processes that enable the socio-technical transformation necessary for deep renovation acceleration.

CRediT authorship contribution statement

Omar Doukari: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Tugce Bacnak:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Elham Mahamedi:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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